

Smart Energy Consumption Analysis and Prediction using Machine Learning with Device-Level Insights

Documentation prepared by: Channa Koushik Kumar

1. Introduction

The rapid increase in electricity demand, coupled with limited visibility into device-level energy usage, has led to significant energy wastage, higher electricity bills and inefficient appliance operation. Conventional electricity billing systems provide only aggregate monthly consumption values, offering little insight into *where*, *when* and *how* energy is consumed.

This project, **Smart Energy Consumption Analysis and Prediction using Machine Learning with Device-Level Insights**, addresses these challenges by analysing detailed, timestamped smart home energy data. The system focuses on understanding device-wise consumption patterns, preparing the data for advanced time-series modelling and laying the foundation for accurate energy forecasting and intelligent energy-saving recommendations.

This document specifically covers **Milestone 1 (Week 1–2)** and demonstrates a clear understanding of data collection, exploration, cleaning, validation and initial exploratory analysis.

2. Project Objectives

The primary objectives for the first two weeks of the project are to understand and structure the Smart Home Energy Monitoring Dataset, ensure high data quality through systematic cleaning and validation, perform exploratory data analysis (EDA) to understand consumption patterns, prepare a timestamped dataset suitable for time-series analysis and establish a strong preprocessing foundation for future machine learning models.

3. Dataset Description

The analysis is based on the Smart Home Energy Consumption Dataset, which contains detailed appliance-level electricity usage recorded over time.

Dataset Characteristics

The dataset consists of **100,000 records** and **8 features**. It is a structured dataset with strong time-series characteristics, making it suitable for temporal analysis and forecasting tasks.

Key Columns

- **Home ID** – Uniquely identifies each household
- **Appliance Type** – Category of appliance (e.g., Air Conditioner, Heater, TV)
- **Energy Consumption (kWh)** – Energy consumed by the appliance measured in kilowatt-hours
- **Time** – Time at which the energy consumption was recorded
- **Date** – Date on which the energy consumption was recorded
- **Outdoor Temperature (°C)** – Ambient outdoor temperature at the time of recording
- **Season** – Seasonal context during which the energy consumption occurred
- **Household Size** – Number of occupants in the household

4. Data Loading and Initial Inspection

The dataset was initially loaded into a pandas DataFrame for structured analysis. An overview of the data was obtained by inspecting the first and last five records.

Activities Performed

The dataset was loaded into a pandas DataFrame, after which the first and last five records were displayed using the head() and tail() functions. Column names and data types were also verified to ensure structural correctness.

Observations

The dataset structure was consistent and well-defined. All expected columns were present and no structural anomalies were observed during the initial inspection.

	Home ID	Appliance Type	Energy Consumption (kWh)	Time	Date	Outdoor Temperature (°C)	Season	Household Size
0	94	Fridge	0.20	21:12	2023-12-02	-1.0	Fall	2
1	435	Oven	0.23	20:11	2023-08-06	31.1	Summer	5
2	466	Dishwasher	0.32	06:39	2023-11-21	21.3	Fall	3
3	496	Heater	3.92	21:56	2023-01-21	-4.2	Winter	1
4	137	Microwave	0.44	04:31	2023-08-26	34.5	Summer	5

Fig 1: First five rows of the dataset

	Home ID	Appliance Type	Energy Consumption (kWh)	Time	Date	Outdoor Temperature (°C)	Season	Household Size
99995	124	Microwave	0.42	09:56	2023-09-28	20.5	Summer	1
99996	184	Computer	0.71	12:48	2023-05-27	-5.4	Spring	2
99997	101	Dishwasher	0.25	05:45	2023-02-18	35.6	Winter	3
99998	423	Air Conditioning	2.69	12:39	2023-04-20	3.7	Spring	1
99999	429	Fridge	0.37	18:46	2023-02-27	36.0	Winter	5

Fig 2: Last five rows of the dataset

5. Data Cleaning and Validation

Ensuring data integrity is critical before performing analysis or building predictive models. Multiple validation checks were conducted.

5.1 Missing Value Analysis

- Checked for null or missing values across all columns
- Result: **No missing values found**

This confirms that the dataset is complete and reliable for further analysis.

5.2 Duplicate Record Handling

- Checked for duplicate rows
- Result: **No duplicate records detected**

This ensures each record represents a unique observation.

5.3 Energy Consumption Validation

- Verified that all values in **Energy Consumption (kWh)** were non-negative
- Rationale: Negative energy values are physically impossible and indicate data errors
- Result: **No invalid or negative values found**

6. Data Transformation: Timestamp Creation

To enable time-series analysis, the separate Date and Time columns were merged into a single Timestamp column.

Transformation Steps

The Date and Time columns were converted into datetime format and combined into a unified Timestamp column. After successful conversion, the original Date and Time columns were removed from the dataset.

Outcome

After transformation, the dataset became fully time-series ready. The presence of a unified Timestamp column enables resampling at hourly, daily, or weekly levels and supports temporal trend analysis and forecasting in later stages of the project.

The transformed dataset was saved as:

timestamped_smart_home_energy_consumption_dataset.csv

	Home ID	Appliance Type	Energy Consumption (kWh)	Outdoor Temperature (°C)	Season	Household Size	Timestamp
0	94	Fridge	0.20	-1.0	Fall	2	2023-12-02 21:12:00
1	435	Oven	0.23	31.1	Summer	5	2023-08-06 20:11:00
2	466	Dishwasher	0.32	21.3	Fall	3	2023-11-21 06:39:00
3	496	Heater	3.92	-4.2	Winter	1	2023-01-21 21:56:00
4	137	Microwave	0.44	34.5	Summer	5	2023-08-26 04:31:00

Fig 3: Dataset with Timestamp column after transformation

7. Exploratory Data Analysis (EDA)

Exploratory analysis was performed to understand consumption behaviour at a high level.

7.1 Appliance Distribution Analysis

A bar chart was used to visualize the frequency of each appliance type in the dataset. This analysis helps identify which appliances appear most frequently and provides an overview of device-level representation.

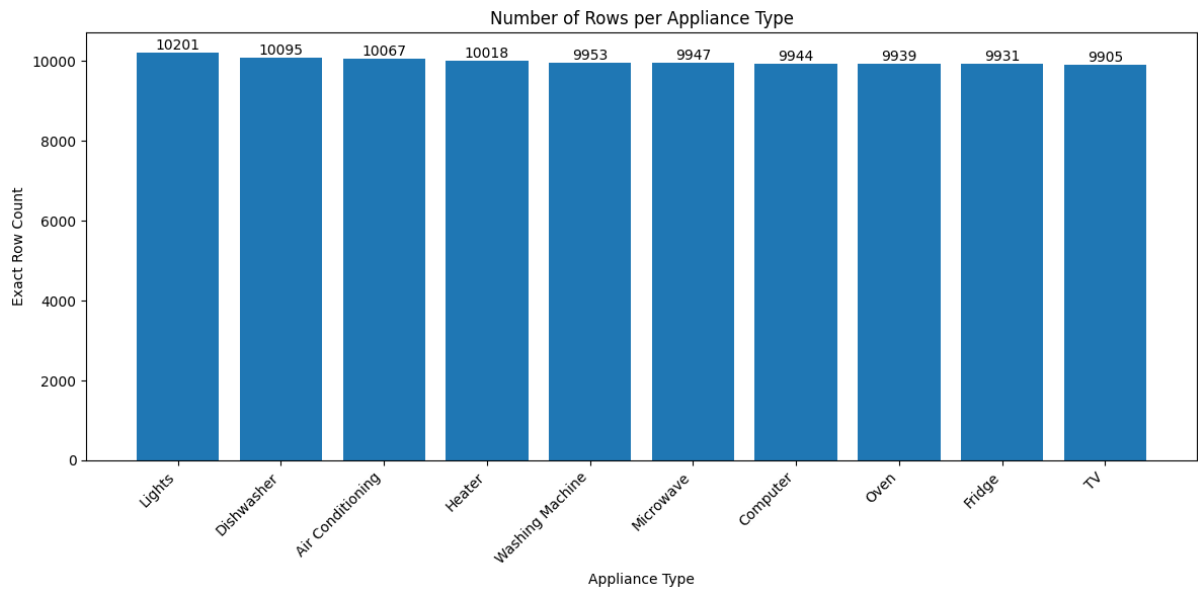


Fig 4: Appliance Type Distribution Bar Chart

7.2 Energy Consumption Distribution

A histogram of Energy Consumption (kWh) was plotted to understand the range, spread and most common consumption values across all appliances.

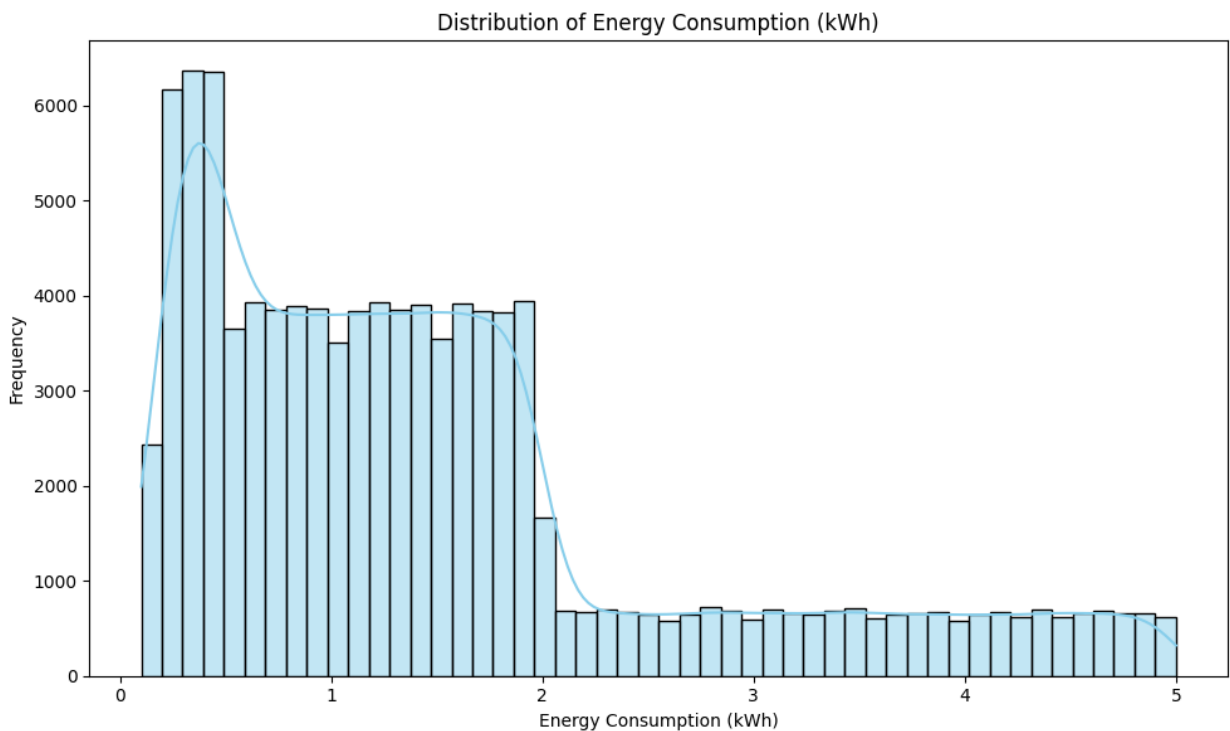


Fig 5: Energy Consumption Histogram

7.3 Appliance-wise Energy Consumption

- Aggregated total energy consumption by appliance type
- Visualized using a bar graph

Key Insights

Air Conditioner and Heater appliances showed the highest total energy consumption, indicating their dominant contribution to overall usage. Appliances such as Dishwasher, Lights, Oven, TV, Microwave and Computer exhibited moderate and comparable usage patterns. The Fridge displayed the lowest energy consumption, which aligns with expectations for an appliance that operates continuously but at low power levels.

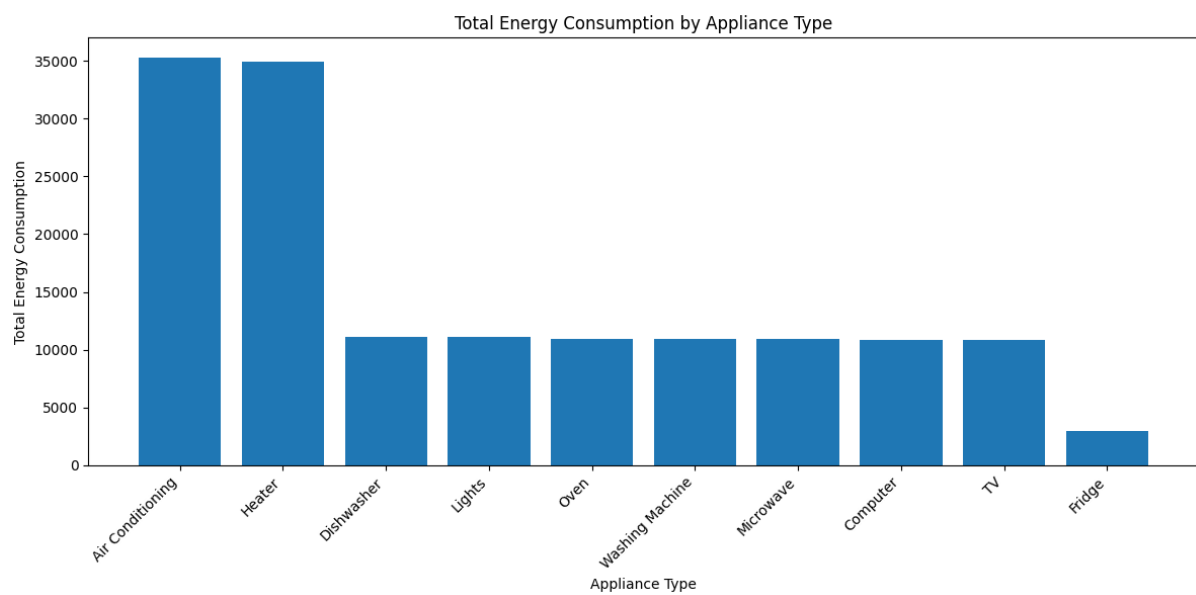


Fig 6: Total Energy Consumption by Appliance

8. Appliance-Level Grouping and Analysis

The dataset was grouped by **Appliance Type** to compute both the total (sum) and average (mean) energy consumption for each appliance category.

Findings

High-consumption appliances were identified as prime candidates for optimization strategies. Appliance-wise aggregation provides actionable insights for demand-side energy management and supports informed decision-making for energy efficiency improvements.

This step establishes the analytical foundation for future modules such as smart suggestions and anomaly detection.

9. Time-Based Energy Consumption Overview

A time-series visualization was created by converting the Timestamp column into datetime format, sorting the records chronologically and aggregating energy consumption daily.

Purpose

This visualization helps identify long-term consumption trends, detect abnormal spikes in usage and observe potential seasonal or cyclical patterns over time.

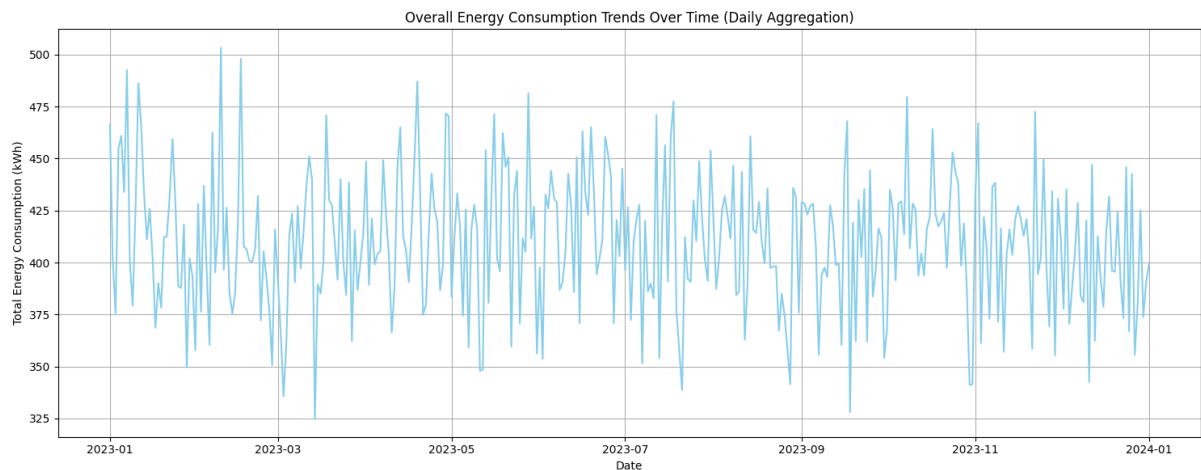


Fig 7: Energy Consumption Line Plot

10. Export of Cleaned Dataset

After cleaning and transformation, the dataset was sorted by Appliance Type and exported as **cleaned_energy_consumption_data.csv**. This dataset serves as the input for feature engineering, machine learning model development and dashboard visualizations.

11. Completed Objectives (Milestone 1)

The dataset was successfully collected and understood, data integrity was validated, timestamp conversion was completed to ensure resampling readiness, exploratory data analysis was performed, and a well-documented preprocessing pipeline was established.

This milestone confirms readiness for **Milestone 2**, which focuses on feature engineering and baseline model development.

12. Key Takeaways and Next Steps

Key Takeaways

The dataset is clean, consistent and ready for analysis. Device-level data provides significantly richer insights compared to traditional billing data and high-energy appliances represent major opportunities for optimization and energy savings.

13. Conclusion

This documentation demonstrates a thorough understanding of smart energy data preprocessing and exploratory analysis. The structured approach ensures a strong technical foundation for machine learning-based energy forecasting and smart home energy optimization in subsequent project phases.